Arshia Akhtarkavan

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Education

2023 – Expected 2027 Bachelor of Science, **University of Arizona**, Tucson, AZ (GPA: 4.0/4.0)
W.A. Franke Honors College
Majors: Physics, Mathematics (Comprehensive Emphasis), & Astronomy

2022 – 2023 Diploma, **Saba International High School**, Frankfurt, Germany

2020 – 2022 High School, **The Atomic Energy Institute**, Tehran, Iran

Research Experience

Undergraduate Researcher in Astrophysics Nov 2023 – Present

Steward Observatory, University of Arizona (Advisor: Prof. Egami)

- Lead a study of galaxy kinematics and morphology using orthogonal JWST/NIRCam grism observations, developing forward models; first-author manuscript in preparation.
- Processed and analyzed JWST/NIRCam imaging and slitless spectroscopy to characterize emission-line galaxies in the SAPPHIRES survey.
- Identified and analyzed transient and variable sources in multi-epoch JWST imaging, contributing to published research on high-redshift transients.
- Designed and trained deep-learning models to help resolve ambiguous redshift solutions in JWST grism spectroscopy.
- Full-time during summer and winter breaks; part-time during the academic year.

Dorrit Hoffleit Research Scholar June – August 2026

Yale University (Advisor: Prof. Geha & Prof. Bellinger)

- Conducted theoretical and observational research on RR Lyrae stars in the Draco dwarf galaxy.
- Developed Bayesian statistical methods to measure stellar radial velocities and estimate distances.
- Applied stellar-evolution modeling to study the physical properties of pulsating stars.

Research in Mathematical Physics October 2024 – January 2026

Mathematics Department, University of Arizona (Advisor: Prof. Cherkis) Islands Project

- Conducted independent research on Nahm equations, constructing new and higher-order solutions
- Rebuilt the Islands Project website to support student engagement in mathematical physics research
- Recipient of RII Undergraduate Research Award supporting this project

Publications

1. Forward-Modeling Galaxy Kinematics and Morphology with Orthogonal JWST/NIRCam Grisms in the SAPPHIRES Early Data Release
Akhtarkavan A., Liu Y., Egami E., Sun F., Zhu Y., et al. (manuscript in preparation).

2. The First Photometric Evidence of a Transient/Variable Source at $z > 5$ with JWST
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2025, *ApJ*, 990, 31; arXiv:2504.17007).

3. Slitless Areal Pure-Parallel HIgh-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field
Sun F., Fudamoto Y., Lin X., Helton J. M., Hsiao T. Y-Y., Egami E., **Akhtarkavan A.**, et al. (*ApJS*, accepted; arXiv:2503.15587).

4. Discovery of 39 Transients/Variables ($z_{\rm host} = 0.437$–5.271) in GOODS-N using JADES and CONGRESS JWST/NIRCam Images
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2024; Transient Name Server AstroNote 264).

Conferences and Workshops

1. **"Forward-Modeling Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
Flash Talk and Poster Presentation, *Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond* Workshop
Space Telescope Science Institute (STScI), Baltimore, MD, August 2026
2. **"Modeling RR Lyrae Spectra in the Draco Dwarf Galaxy"**
Contributed Talk, *Yale/CeNAM Nuclear Asteroseismology Workshop*
Yale University, New Haven, CT, July 2026
3. **"Modeling RR Lyrae Spectra in the Draco Dwarf Galaxy"**
Contributed Talk, *Summer Research Symposium*
Yale University, New Haven, CT, July 2026
4. **"Islands Project and the Nahm Equations"**
Talk, *Third Annual Cross-Campus Undergraduate Research Forum and Questionarium*
University of Arizona, Tucson, AZ, April 2026
5. **"Forward-Modeling Emission-Line Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
Contributed Talk, *Steward Observatory Internal Symposium*
Steward Observatory, University of Arizona, Tucson, AZ, March 2026
6. *Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era*
Arizona Cosmology Lab, University of Arizona, Tucson, AZ, January 2026
7. **"Forward-Modeling Emission Line Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
iPoster presentation, *247th Meeting of the American Astronomical Society (AAS 247)*
Phoenix, AZ, January 2026

Awards and Scholarships

Dorrit Hoffleit Research Scholarship, Yale University (2026)
Evelyn O. Bychinsky Promising Astronomer Award (2026)
William F. Lucas Astronomy Scholarship, Steward Observatory (2026)
College of Science Galileo Circle Scholar (Mathematics), University of Arizona (2026)
College of Science Galileo Circle Scholar (Physics), University of Arizona (2026)
Weaver Award for Undergraduate Research (2026)
Kenneth S. Krane Physics Scholarship (2025)
Evelyn O. Bychinsky Promising Astronomer Award (2025)
RII-Sponsored Campuswide Undergraduate Student-Initiated Original Research (2025)
Glenn C. Purviance Scholarship in Physics (2023)
Academic Year Highest Academic Distinction (2023 – Present)
Dean's List with Distinction - Full Time (2023 – Present)
Global Wildcat Tuition Award, University of Arizona (2023 – Present)
Silver Medal, National Astronomy and Astrophysics Olympiad (2022)

Independent Studies

Quantum Field Theory Fall 2026, in progress

Physics Department, University of Arizona (Advisor: Prof. Meinel)

– Graduate-level independent study

– Completing weekly problem sets and a final-term project

Nahm Equations and Mathematical Physics Fall 2025

Mathematics Department, University of Arizona (Advisor: Prof. Cherkis)

– Studied Nahm equations and related topics in gauge theory and representation theory

– Wrote a final report on the construction of higher-order Nahm solutions

Professional Experience

Astronomical Software Engineering Intern Aug 2026 – Present

TIMESTEP, University of Arizona (Event Horizon Telescope Group; Supervisor: Dr. CK Chan)

– Participating in an academic-year internship focused on developing `Hallmark`, a lightweight Python package for version-controlling and distributing scientific data products across Event Horizon Telescope analysis pipelines.

Mechanical Engineering Workshop Intern Jan – Mar 2023

FH Aachen University, Essen, Germany

– Gained hands-on experience operating machining and workshop equipment and interpreting and producing technical schematics.

Teaching and Outreach

Mathematics Grader Aug 2025 – May 2026

Mathematics Department, University of Arizona, Tucson, AZ

– Graded assignments and examinations, prepared detailed solutions, and provided written feedback for upper-division courses in differential equations and linear algebra.

Undergraduate Mentor Oct 2026 – Present

University of Arizona, Tucson, AZ

– Participate in an undergraduate mentoring program, providing peer guidance and support to one student.

Astronomy Olympiad Teaching Assistant Oct 2022 – Aug 2023

The Atomic Energy Institute, Tehran, Iran

– Taught cosmology and prepared lecture notes for Astronomy Olympiad students, based on textbooks by Ryden and Carroll.

– Selected problems and led problem-solving sessions in celestial mechanics.

Educational Materials Translator 2025

Arizona Astrobiology Center, University of Arizona, Tucson, AZ

– Translated Astrobiology lecture materials from English into Farsi for Persian-speaking elementary school students

Skills

Programming Languages	Python, C/C++, SQL
Scientific & Data Analysis Libraries (Python)	NumPy, Matplotlib, PyTorch, Astropy, SciPy, Specutils, Photutils, Pandas, JAX
Astronomy & Astrophysics Software	DS9, JWST Pipeline, Source detection and Photometry, Webbpsf, NIRCam imaging and grism spectroscopy
Software & Research Systems	Linux/Unix, Git/GitHub, HPC environments, remote computing (ssh), GPU-accelerated computing (CUDA), Jupyter, VS Code, Emacs, LaTeX, Microsoft Office
Concepts	Data Processing Pipelines, Scientific Computing, Machine Learning, Numerical Methods

Languages

English	Fluent
Farsi (Persian)	Native
German	B1 level
Mandarin Chinese	Currently learning